2025-26 Archimedean Satellite Invitational Meteorology
Answer Key

1. A, B
2. D
3. B
4. C
5. D
6. B
7. Liquid water and ice can both exist at 0 C. For liquid water to turn into ice, the intermolecular bonds must form and strengthen to become ice, a process that releases heat. [1] This heat is not associated with a temperature change of the water and is called latent heat. [1]
8. D
9. Increases intensity of seasons
10. closer, farther
11. formation, dissipation
12. CCW, CCW, Coriolis
13. Katabatic, denser, cause
14. Day, night
15. C
16. C
17. B
18. A decrease in temperature decreases the amount of water vapor that the air can hold [1], and relative humidity is defined as the amount of water vapor in the air out of the amount it can hold [1]. Thus, even though the amount of water vapor stays the same, the amount that can be held decreases, increasing relative humidity [2].

19. Lifted Condensation Level
20. Any two of the following: Orographic lift (pushed into a mountain or other form of rising terrain), convergent lift (two air masses collide and go up), convective (less dense air rising due to buoyancy), frontal (air masses rising along a front). Or any other with justification.

21. Ac
22. Ci
23. Cs
24. Ln
25. Cb
26. Cc
27. As
28. Ns
29. St
30. PCb
31. Rainbow
32. Sundogs
33. Green Flash
34. Crepuscular Rays
35. Mirage
36. Rainbow
37. Halo
38. Mirage
39. Barometer
40. Doppler Radar
41. Radiosonde/Rawinsonde
42. Anemometer
43. Rain Gauge
44. Wind Profiler
45. Sling Psychrometer
46. Snowboard
47. Occluded
48. 1028
49. 3.1e-3 P/m [1]. The pressure over SD is 1031 mb = 103100 Pa [1] and Wisconsin is 996 mb = 99600 Pa [1]. The pressure difference is 3500. The distance, in meters, is 700 miles * 1609 meters / mile = 1126300 meters [1]. The pressure gradient is defined as delta P / D which is 3500 / 1126300 = 3.1e-3 P/m.

50. Closer to WI [1]. Isobars are much more tightly packed near the low pressure cell, so pressure is changing more rapidly with distance, causing a steeper pressure gradient [1].
51. There will be little to none due to high pressure inhibiting convection [2], but some may form such as small cumulus/stratocumulus clouds and some cirrus clouds.
52. GA-FL area. [1] Around a low pressure cell, winds will flow CCW [2], so winds are flowing into this region with the cold front, showing strong cold air advection. Other locations with accurate justification will be accepted.
53. New York
54. Intensifying. [1] The warm front is very small and the state of the occluded front is characteristic of a growing cyclone. [1]
55. cP from central Canada [1], mP from Atlantic ocean, fueling warm front [1], mT from Atlantic ocean, part of cold front [1]
56. Pressure trough
57. Dry line
58. Meridional
59. East coast but mostly NE
60. Under a trough
61. Trough: over ID, MT, UT, AZ [1], positively tilted [1], over Atlantic Ocean/NE US [1], negatively tilted [1]. Ridge: over ND, SD, NE, KS, OK [1], positively tilted [1].
62. New York area [1]. Cyclonic curvature is strong and the flow is carrying higher vorticity air toward the East Coast, favoring ascent and supporting the surface low there [2].
63. A: Rain, B: Snow, C: Sleet, D: Freezing Rain
64. 12.8 C
65. 43 F
66. 100%
67. 74.8 mph
68. 996 mb
69. 998 mb
70. Cb
71. Ci
72. Moderate
73. 30th, 16:51
74. 220 degrees (SW), sustained 20.7 mph, gusts 32.2 mph (either will be accepted)
75. 10 statute miles
76. Broken: 2000 ft. Overcast: 7000 ft.
77. Temperature: 12 C. Dew Point: 9 C
78. 29.92 inHg
79. 1013.4 hPa